

Homework 5

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Problem 1

Part a

The MMSE one-step-ahead forecast is:

$$\hat{x}_{t+1|t} = E[x_{t+1} \mid w_t, w_{t-1}, \dots].$$

$$x_{t+1} = w_{t+1} + \theta w_t.$$

Using conditional expectation:

$$\hat{x}_{t+1|t} = E[w_{t+1} \mid w_t, w_{t-1}, \dots] + \theta E[w_t \mid w_t, w_{t-1}, \dots] = \theta w_t.$$

The forecast error is:

$$e_{t+1} = x_{t+1} - \hat{x}_{t+1|t} = w_{t+1}.$$

The MSE is:

$$E[e_{t+1}^2] = E[w_{t+1}^2] = \sigma_w^2.$$

Part b

The truncated forecast is given by:

$$\tilde{x}_{n+1|n} = w_n + \theta w_{n-1}.$$

$$x_{n+1} = w_{n+1} + \theta w_n.$$

The forecast error is:

$$e_{n+1} = x_{n+1} - \tilde{x}_{n+1|n} = (w_{n+1} + \theta w_n) - (w_n + \theta w_{n-1}).$$

$$e_{n+1} = w_{n+1} - w_n - \theta w_{n-1}.$$

The MSE is:

$$E[e_{n+1}^2] = E[w_{n+1}^2] + E[w_n^2] + \theta^2 E[w_{n-1}^2] = \sigma_w^2 + \sigma_w^2 + \theta^2 \sigma_w^2 = \sigma_w^2(1 + \theta^2 + 2n).$$

Comparison

As $n \rightarrow \infty$, the truncated forecast error variance grows linearly, indicating poor performance for large n .

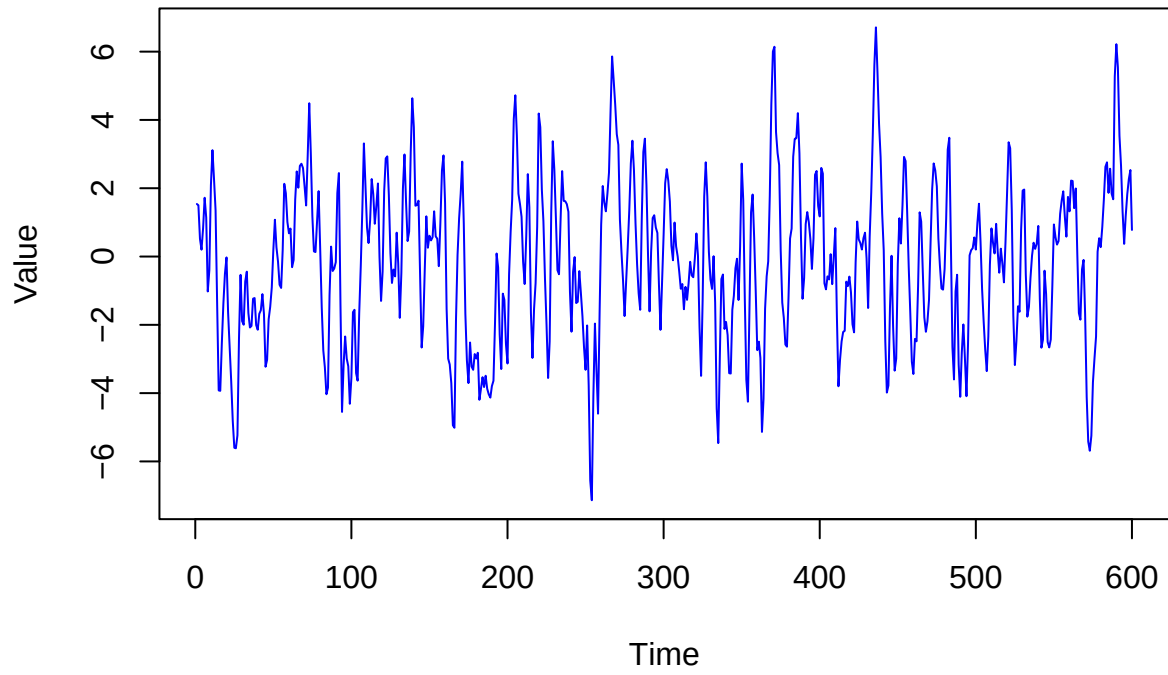
Summary

```
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1       1.4868 0.0403 36.8615 0.0000
## ar2      -1.0876 0.0692 -15.7180 0.0000
## ar3       0.5692 0.0691  8.2316 0.0000
## ar4      -0.1568 0.0403 -3.8886 0.0001
## xmean    -0.1779 0.2182 -0.8153 0.4152
##
## sigma^2 estimated as 1.020513 on 595 degrees of freedom
##
## AIC = 2.881961   AICc = 2.882129   BIC = 2.92593
##
##
##           Length Class Mode
## fit             14     Arima list
## degrees_of_freedom 1     -none- numeric
## ttabel           20     -none- numeric
## ICs              3     -none- numeric
```

Looking at the plot of X_t against time for the first series, it does not appear that the mean and variance change with time, so no transformations or differencing are needed. Looking at the ACF and PACF, my initial guess for the model is an AR(4) because the PACF has 4 terms that appear significant then it drops almost all the way to zero while the ACF tails off. Based on the diagnostic plots for the AR(4) that was fit, it seems to be satisfactory because the standardized residuals appear to be stationary, the ACF of the residuals shows no significant lags, the normal QQ plot of the residuals shows that they do not have abnormally light or heavy tails, and the LBP shows that there are no lags that demonstrate significant cumulative autocorrelation of residuals at any lag up to 20, so the final model is ARIMA(4,0,0). The `xmean` term in the model summary is the only one that is not significant at the $p < 0.01$ level, which is because the data is already very close to mean 0.

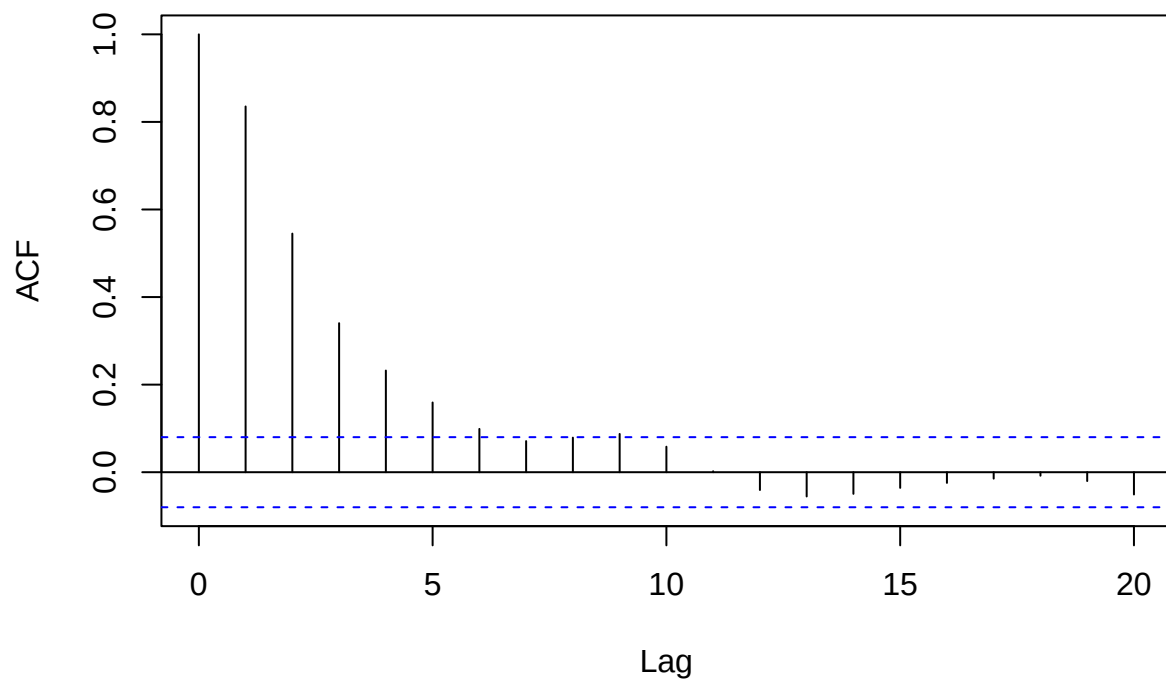
Initial series plot:

Time Series – dat1

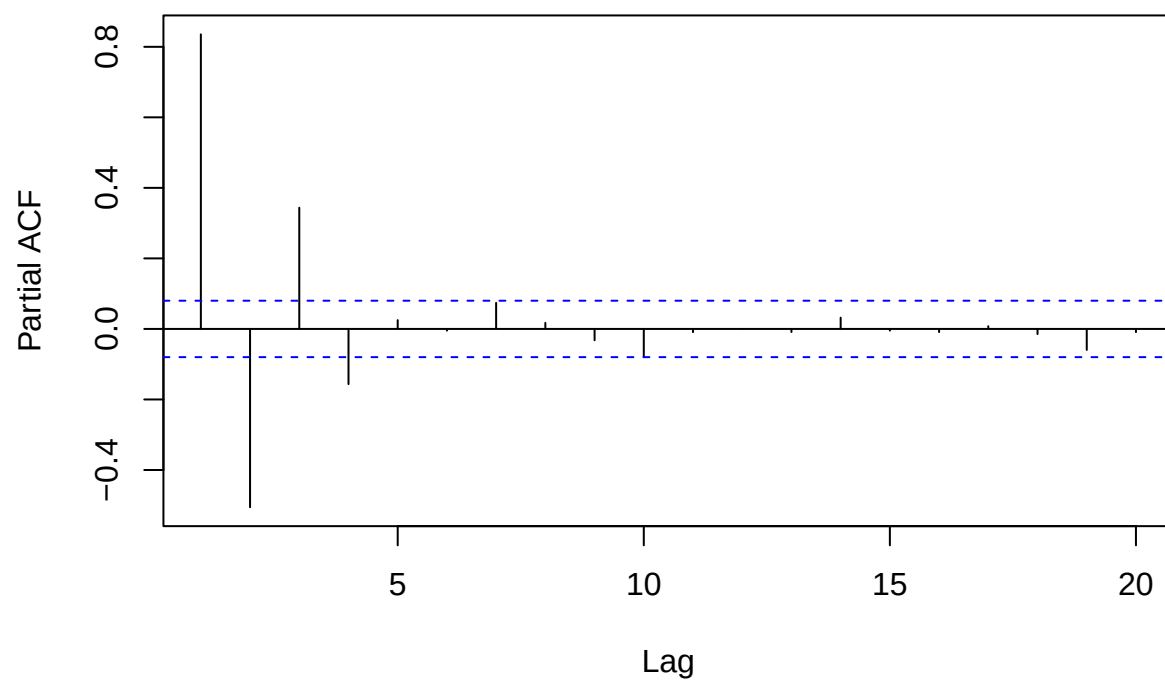


ACF and PACF:

ACF of dat1



PACF of dat1



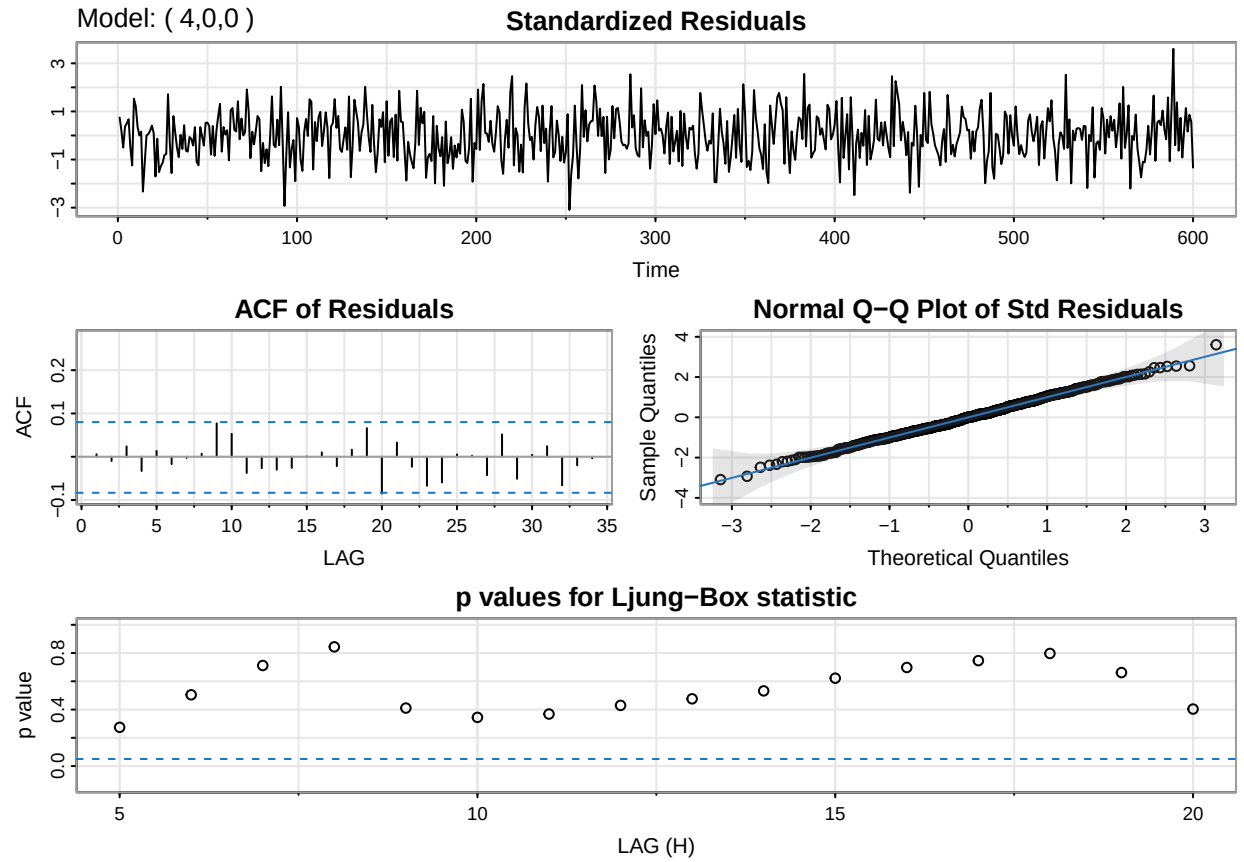
Diagnostic plots for AR(4) fit:

```
## initial  value 0.836111
## iter   2 value 0.621847
## iter   3 value 0.416015
## iter   4 value 0.256297
## iter   5 value 0.226524
## iter   6 value 0.101954
## iter   7 value 0.078047
## iter   8 value 0.033464
## iter   9 value 0.020111
## iter  10 value 0.012826
## iter  11 value 0.012677
## iter  12 value 0.012674
## iter  13 value 0.012674
## iter  14 value 0.012673
## iter  15 value 0.012673
## iter  16 value 0.012673
## iter  17 value 0.012673
## iter  18 value 0.012673
## iter  19 value 0.012673
## iter  20 value 0.012673
## iter  21 value 0.012673
## iter  22 value 0.012673
## iter  22 value 0.012673
## iter  22 value 0.012673
```

```

## final value 0.012673
## converged
## initial value 0.012048
## iter 2 value 0.012046
## iter 3 value 0.012045
## iter 4 value 0.012045
## iter 5 value 0.012044
## iter 6 value 0.012043
## iter 7 value 0.012042
## iter 8 value 0.012042
## iter 9 value 0.012042
## iter 10 value 0.012042
## iter 10 value 0.012042
## iter 10 value 0.012042
## final value 0.012042
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      1.4868 0.0403 36.8615 0.0000
## ar2     -1.0876 0.0692 -15.7180 0.0000
## ar3      0.5692 0.0691  8.2316 0.0000
## ar4     -0.1568 0.0403 -3.8886 0.0001
## xmean   -0.1779 0.2182 -0.8153 0.4152
##
## sigma^2 estimated as 1.020513 on 595 degrees of freedom
##
## AIC = 2.881961 AICc = 2.882129 BIC = 2.92593
##

```



Problem 3

Summary

The model summarized below is ARIMA(3,1,0).

```
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE  t.value p.value
## ar1           0.5903 0.0409  14.4438  0.0000
## ar2          -0.5063 0.0427 -11.8604  0.0000
## ar3           0.0538 0.0408   1.3194  0.1876
## constant    -0.0011 0.0508  -0.0218  0.9826
##
## sigma^2 estimated as 1.146769 on 595 degrees of freedom
##
## AIC = 2.992656  AICc = 2.992769  BIC = 3.029344
##

##           Length Class  Mode
## fit           14      Arima list
## degrees_of_freedom 1      -none- numeric
## ttable         16      -none- numeric
## ICs             3      -none- numeric
```

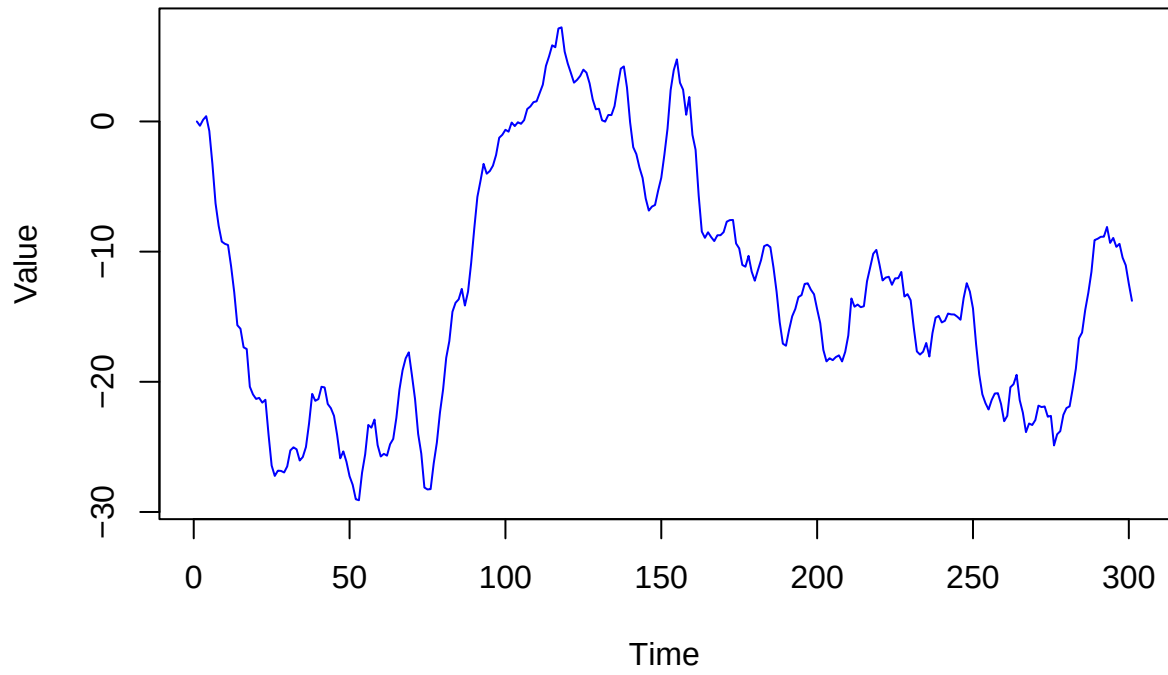
Explanation

Looking at the plot of X_t against time for the second series, it looks similar to a random walk which is not stationary so we have to take the difference to make it stationary. After taking the first difference it does look stationary so no more differencing is needed. Looking at the ACF and PACF after differencing once, my initial guess for the model is an AR(3) because the PACF has 3 terms that appear significant then it drops almost all the way to zero while the ACF tails off. Based on the diagnostic plots for the AR(3) that was fit on the differenced series, it seems to be satisfactory because the standardized residuals appear to be stationary, the ACF of the residuals shows no significant lags, the normal QQ plot of the residuals shows a slightly heavier left tail and a slightly lighter right tail than expected but nothing out of the ordinary, and the LBP shows that there are no lags that demonstrate significant cumulative autocorrelation of residuals at any lag up to 20, so the final model is ARIMA(3,1,0). The xmean term in the model summary is the only one that is not significant at the $p < 0.01$ level, which is because the data is already very close to mean 0.

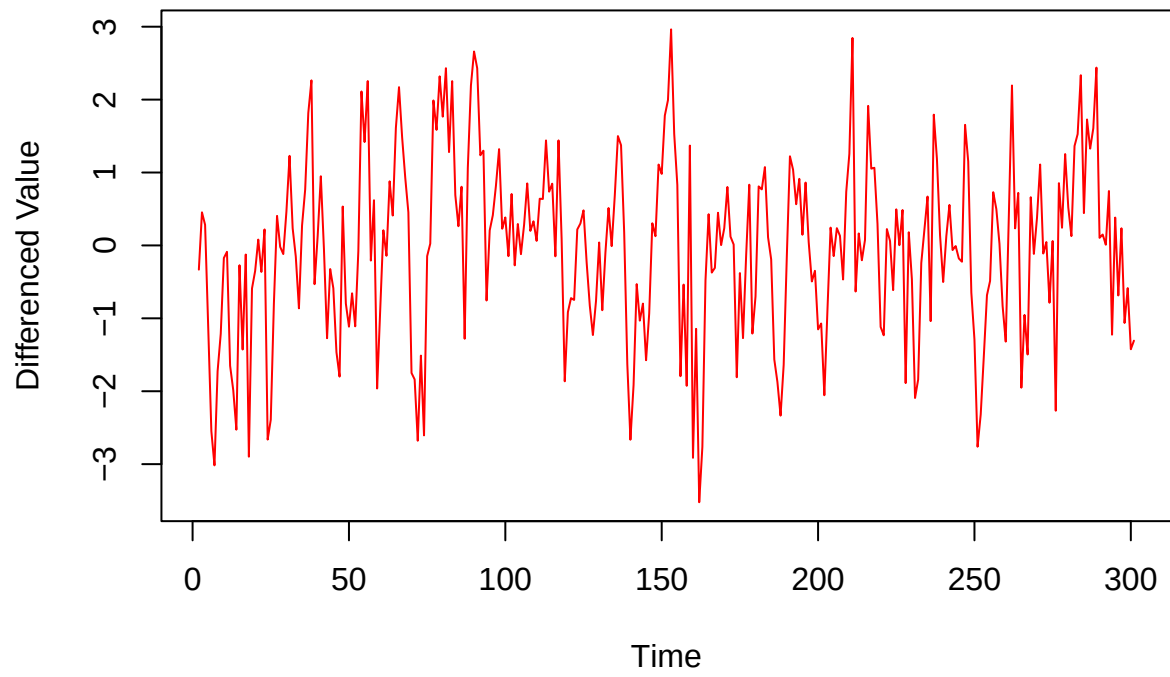
Output

Initial series plots with and without differencing:

Time Series – dat2 (Original)

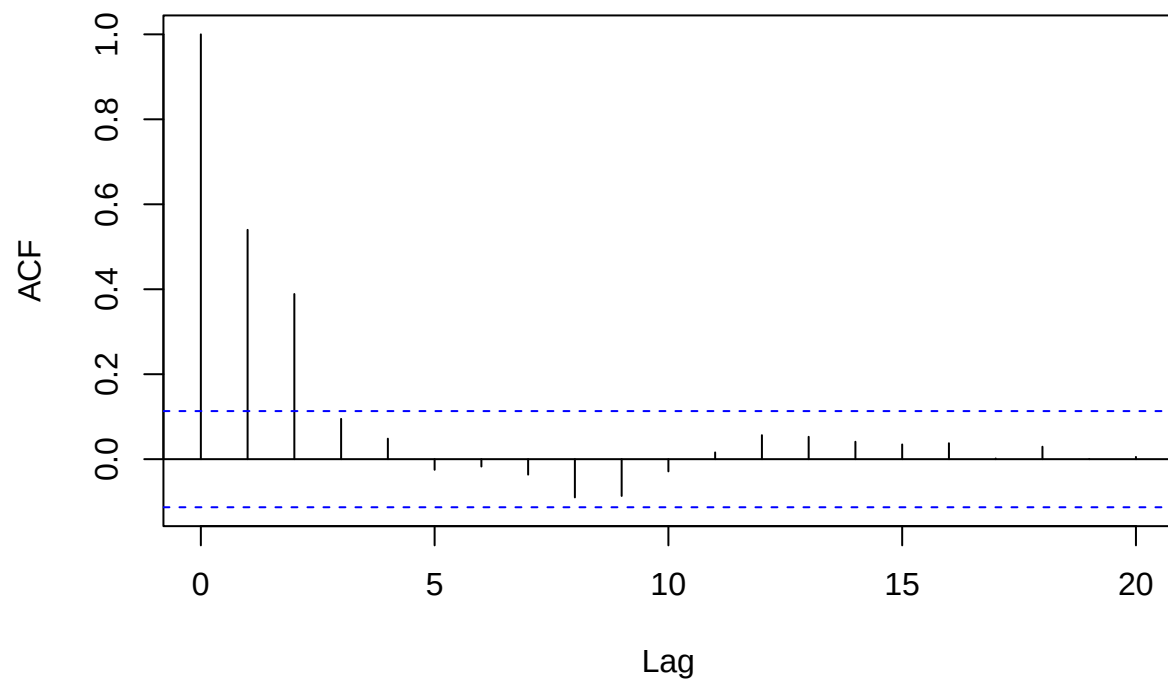


Time Series – dat2 (Differenced)

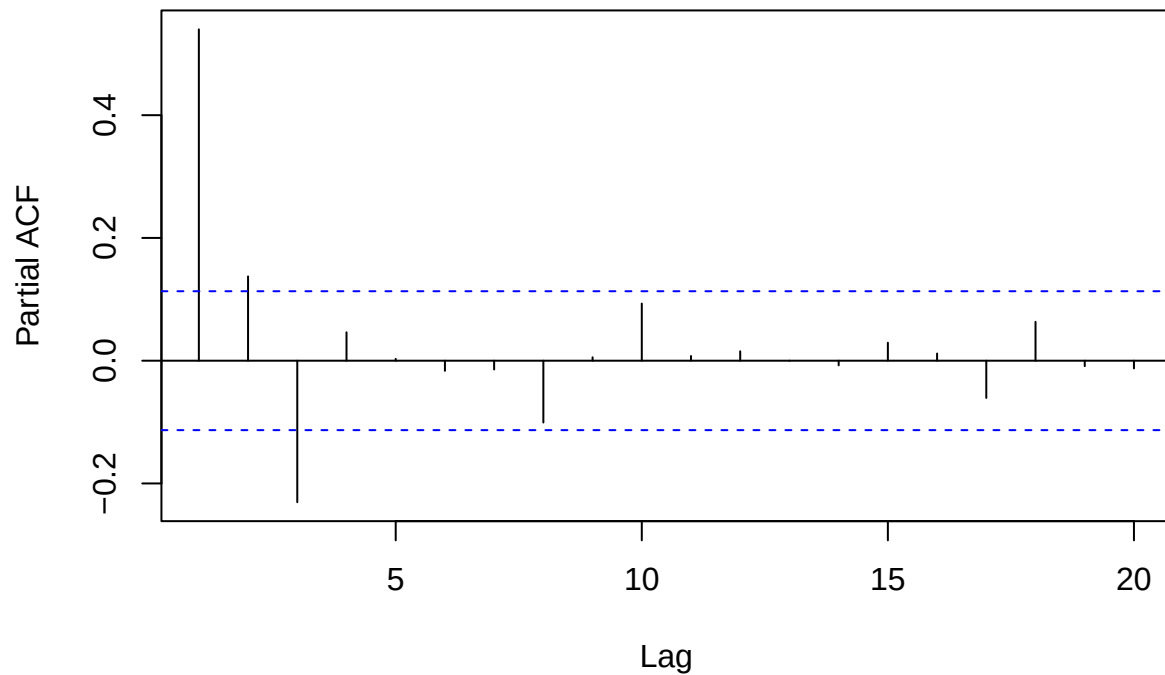


ACF and PACF:

ACF of dat2 after differencing



PACF of dat2 after differencing



Diagnostic plots for AR(3) fit of the differenced series:

```
## initial value 0.204783
## iter 2 value 0.123831
## iter 3 value -0.002884
## iter 4 value -0.005857
## iter 5 value -0.006045
## iter 6 value -0.006052
## iter 7 value -0.006054
## iter 8 value -0.006054
## iter 9 value -0.006055
## iter 10 value -0.006055
## iter 11 value -0.006055
## iter 12 value -0.006055
## iter 13 value -0.006055
## iter 13 value -0.006055
## iter 13 value -0.006055
## final value -0.006055
## converged
## initial value -0.009355
## iter 2 value -0.009359
## iter 3 value -0.009360
## iter 4 value -0.009360
## iter 5 value -0.009360
## iter 6 value -0.009360
## iter 7 value -0.009360
```

```
## iter    8 value -0.009360
## iter    8 value -0.009360
## iter    8 value -0.009360
## final   value -0.009360
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4975 0.0560  8.8830 0.0000
## ar2      0.2460 0.0614  4.0096 0.0001
## ar3     -0.2307 0.0561 -4.1163 0.0000
## xmean   -0.0511 0.1171 -0.4364 0.6629
##
## sigma^2 estimated as 0.9796619 on 296 degrees of freedom
##
## AIC = 2.85249  AICc = 2.852942  BIC = 2.91422
##
```

